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Bio-optical studies on chlorophyll-a concentration in Hooghly River, India

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ABSTRACT

The concentration of chlorophyll-a in the aquatic environment is a good indicator to understand its tropic status and also used to monitor the biogeochemical parameter. The aim of this study is to identify the spectral signature of Hooghly River, India using Satlantic hyperspectral ocean colour radiometer and developing the regional algorithm for retrieval for chlorophyll-a concentration. This approach is based on the development of an empirical band ratio algorithm. The calibrated results between the in-situ chlorophyll-a and in-situ remote sensing reflectance show a good correlation with $R^2 = 0.77$. In-situ and satellite validation also gives better accuracy ($R^2 = 0.78 \& 0.73$) in the estimation of chlorophyll-a concentration. So the satellite-based approach provides a good correlation with in-situ data. This study was helped in monitoring and retrieval of chl-a concentration in the Hooghly River, Kolkata.

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1. Introduction

All manmade activates impact the river and specifically, water quality is significantly affected [1]. The Hooghly River is the major source of water to the West Bengal state. Therefore monitoring the water quality parameter was mandatory to track the contamination zone in the river. The field-based analysis of water quality is always expensive and time-consuming [2]. So the satellite-based retrieval of water quality parameter is highly recommended for monitoring the aquatic environment. The majority of remote sensing sensors are not suitable for water quality monitoring. There are very few open access sensors with adequate spatial and temporal resolution is available. Among them, Sentinel-2 Multispectral Instruments (MSI) has a high spatial resolution ranging from 10 m to 60 m and 5 days revisit time. Even though the Sentinel-2 MSI data belongs to a land-based monitoring sensor, there was several research works carried out in estuarine and coastal water [3]. For Sentinel-2 MSI data, several atmospheric correction methods have been developed. Case-2 Regional Coastal Colour (C2RCC) is an atmospheric correction tool for retrieving Bottom-of-Atmosphere data (level-2) from Top of Atmosphere data (level-1C). The Sentinel-2 MSI bands are suitable for developing the regional algorithm and estimation of optically active substances in the inland and coastal water. The chlorophyll-a concentration

is calculated using Sentinel-2 MSI data by Grendaite [4]. The developed algorithm is well-calibrated and validated using in-situ data in the Barra-Bonita reservoir, Brazil. The investigation applies Sentinel-2 MSI bands for estimation of optically active substances and the developed regional algorithm. The accuracy of the Chl a retrieval algorithm for turbid water is evaluated by Sentinel-2 data. Commonly used band ratio algorithm (green to red) for estimating different water constituent in Estonian lakes was investigated [5]. The regional algorithm is developed based on in-situ and lab measurements. The in-situ measurements must be coinciding with satellite pass time within 3 h. The distribution pattern of chlorophyll-a varies with river flow. In satellite remote sensing, cloud cover is a major drawback for the retrieval of chlorophyll-a concentration [6]. So the in-situ measurements should be done on cloud-free days. The measured in-situ water constituents are validated with Sentinel-2 MSI data for retrieval of chlorophyll-a concentration in Hooghly River with high spatiotemporal resolution [7].

2. Materials and methods

2.1. Study area

The study area is situated near the Nayachar Island and the geographical location between the latitudes of 22.01° to 21.75° N and longitudes of 88.01° to 88.08° E in West Bengal, India. The Ganga

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River originates in the Himalayan foothills and flows through the state of West Bengal. The river then passes through several districts in West Bengal and finally reaches the Bay of Bengal [8]. The major cities called Kolkata and Howrah are located on the bank of the Hooghly River. Mostly all the river in the West Bengal is interlinked with Hooghly River (Fig. 1). The Ecosystem of the Hooghly River is under stress due to the international ship route, industrial activities, shipping harbour, agricultural lands and households. Additional to that, ferry transportation places a major role in the disturbance of the aquatic environment.

2.2. Field measurements

2.2.1. Approach description

The first and foremost field survey is carried on the date of Sentinel-2 MSI satellite pass for remote sensing application. To validate with satellite remote sensing, all sampling points are fixed with an interval of 20 m distance. Simultaneously both the Satlantic Hyperspectral ocean colour radiometric (HyperOCR) data and water samples were collected from each station. The HyperOCR instrument was used for measuring the optical properties of water. To minimize the field error, all the measurements were done only after the boat movements are stabilized. In optical remote sensing, measurements should be carried only in the shadow-free region to avoid the noise in data. At the time of sample acquisition, pitching, rolling, wind and wave action are noticeable not affecting the optical signal [9]. The field measurements are taken from the northern part of the Hooghly River and end up near Nayachar Island on the same day of Sentinel-2 MSI overpass. The total number of 40 samples was collected in April (2018 & 2019) [10].

2.2.2. Water samples

The water samples are collected at one meter depth using a Niskin water sampler, to calibrate and validate the in-situ data with Sentinel-2 satellite data. All the water samples are filtered on the same day by 47 mm Whatman GF/F fibre glass filters of pore size of 0.7 μm with low vacuum pressure (0.2 bar). After filtration, the filter paper was placed in the cryotube and store in the deep freezer for further analysis [11]. The filter papers are added with 10 ml of 90% of acetone and stored in the refrigerator at 4 $^{\circ}\text{C}$ for 24 h. Now the filtrates are extracted by using polytetrafluoroethylene (PTFE) syringe filters of pore size 0.2 μm with 25 mm diameter. The UV-Vis spectrophotometric (Shinaszu-UV 1800) methods are used in a laboratory to measure the chlorophyll-a concentration. The chlorophyll-a concentration is calculated by measuring the optical density at a selected wavelength [12].

Chlorophylla concentration

$$= \frac{[(11.85 * OD664) - (1.548 * OD647) + (0.08 * OD630)]}{v} \quad (1)$$

OD-Optical Density, V- Volume of acetone, v-volume of filtered water

2.2.3. Satlantic radiometer measurements

The remote sensing reflectance (R_{rs}) is measured by calculating the ratio of water-leaving radiance (L_w) and downwelling irradiance (E_d) just above the sea surface. The downwelling irradiance is measured by collecting the sun radiance. The radiance sensor is facing 45 $^{\circ}$ upward and downward to calculate total water-leaving radiance (L_t) and sky radiance (L_s). The water-leaving radiance is calculated by equation (2)

$$L_w() = L_t - F_r \times L_s \quad (2)$$

$$\text{RemoteSensingReflectance}R_{rs}(\lambda) = \frac{L_w(\lambda)}{E_d(\lambda)} \quad (3)$$

Fresnel Reflectance- $F_r(\lambda) = 0.028$, L_t - Total water-leaving radiance, L_s - Sky radiance.

2.3. Chl-a empirical model

In ocean colour remote sensing, blue-green band ratio algorithms (60) are commonly used for retrieval of chlorophyll-a concentration in the case 1 water (Open Ocean). In open ocean water, phytoplankton absorption has a significant impact on blue-green band. Whereas in case 2 water (inland and coastal water), the blue-green band is associated with absorption of coloured dissolved organic matter and scattering of total suspended matter [13]. In estuarine and coastal water, the band ratios of 705/560 and 705/665 are often used to retrieve chl-a concentrations, while CDOM absorption is negligible. The semi-empirical band ratio (2B) algorithms used to build a regional algorithm for retrieving chlorophyll-a concentration. The central wavelength of Sentinel 2 data is used in the band ratio algorithm [14,15]. The datasets are fit with polynomial regression for retrieval of chl-a concentration, among other regression analysis.

$$2B = (R_{rs} 705 / R_{rs} 560) \quad (4)$$

$$\text{Chl-a} = a (2B^2) + b (2B) + c \text{ (polynomial constant)} \quad (5)$$

2.4. Sentinel-2 satellite data

The Sentinel 2 MSI (Multispectral Instrument) contains 13 spectral bands which belong to the visible and NIR region with a wide

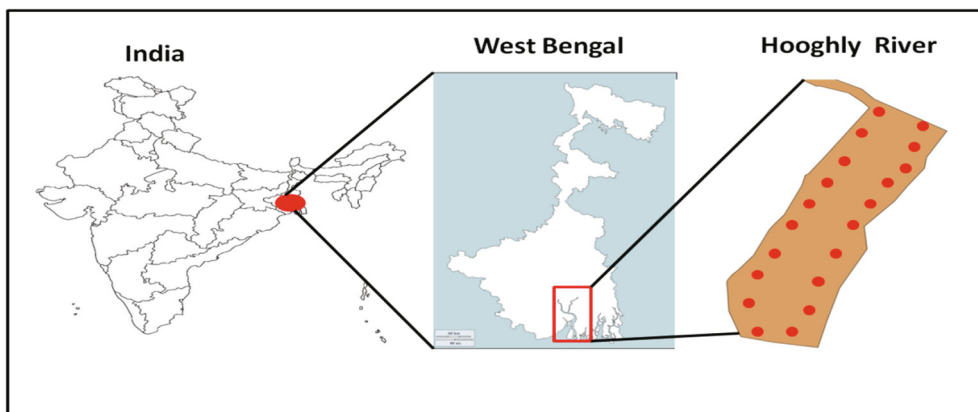


Fig. 1. Study area along with sampling location along the Hooghly River.

field of view (290 km) [16]. Sentinel-2 data has a spatial resolution of 10 m, 20 m and 60 m. All the satellite data are downloaded from the USGS Earth Explorer (<https://earthexplorer.usgs.gov/>) which provides users to query, search and order satellite data for research activities. The downloaded Sentinel 2 level 1C data coincides with survey timing. Sentinel Application Platform (SNAP) software is used to process the Sentinel 2 data from level 1C to level 2 data [17]. Atmospheric correction was applied using a C2RCC processor (Case-2 Regional Coast Colour). The C2RCC is an atmospheric correction tool validated with various inland and coastal studies and is free downloadable from the ESA Sentinel toolbox in SNAP software [18]. All spectral bands were resampled to 20 m resolution and each in-situ data point was covered by one satellite pixel on the average. After atmospheric correction, retrieved remote sensing reflectance is used for satellite validation [19,20].

2.5. Statistical analysis

Performance of the developed regional model was done with statistical analysis.

$$RMSE = \left[\frac{1}{N} \sum (forecast - actual)^2 \right]^{1/2} \quad (6)$$

$$MAPE = (1/N) * \sum (|actual - forecast| / |actual|) * 100 \quad (7)$$

$$\text{Standard Deviation (S.D)} = \sigma = \sqrt{\sum (x - \bar{x})^2 / N} \quad (8)$$

$$\text{Co-efficient of variation (CV)} = \sigma / \mu \quad (9)$$

where, Actual - In situ data, Forecast - Developed data, σ - Standard Deviation, μ - Mean, N- No. of sample

3. Results and discussions

3.1. Statistical analysis of chl-a concentration

Among the 40 datasets, 65% of the datasets are used for calibration and 35% of datasets are used for in-situ validation. Sentinel-2 MSI satellite data (N = 40) were used for satellite validation. The concentration of in-situ chl-a range from 17.0 mg/m³ to 26.7 mg/m³, while satellite-derived chl-a concentration varies from 18.4 to 25.0 mg/m³. Despite the fact that suspended matter predominated in the study area, chl-a concentrations were also present due to anthropogenic activities and municipal wastewater mix. The statistical description of chl-a concentration is given in Table 1.

Where, S.D- Standard Deviation, C.V- coefficient of variance

3.2. Reflectance spectra

In reflectance band, chl-a concentration of case 1 water belongs to the blue and green region. However, due to the complexity of different constituent in case II water, identification of chl-a concentration required spectra signature of water. The study area is the domination of total suspended matter (TSM), coloured dissolved organic matter (CDOM) and chlorophyll-a concentration (Chl-a). As a result, estimating the chl-a concentration in the highly turbid water is more complex in nature. According to Moses [20], the

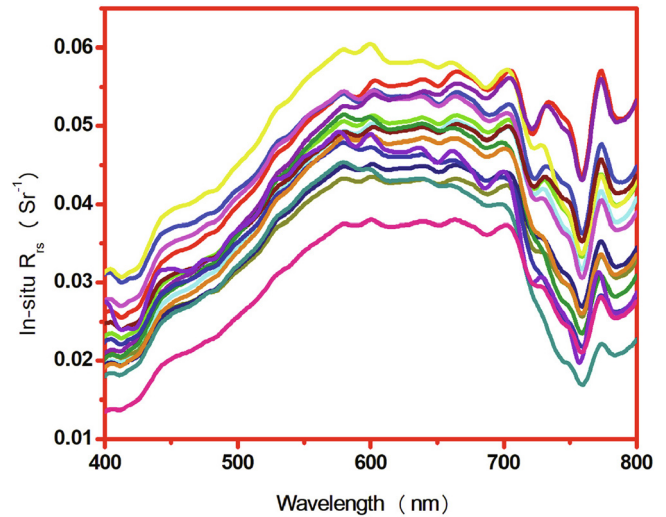


Fig. 2. Remote sensing reflectance spectra using Satlantic HyperOCR Radiometer.

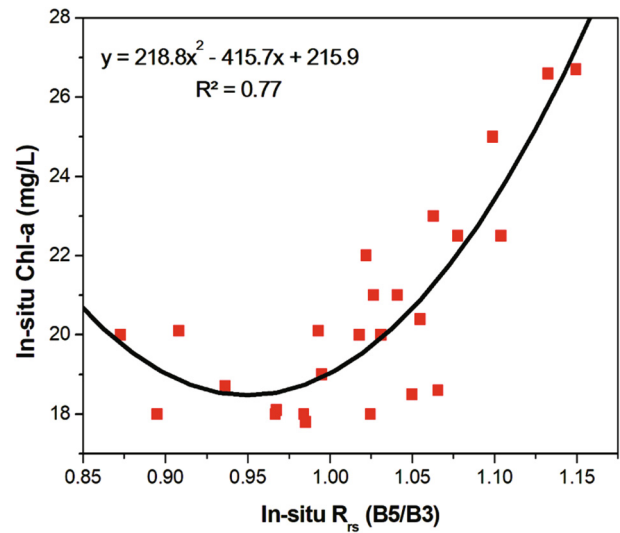


Fig. 3. Two band ratio algorithm between in-situ remote sensing reflectance (B5/B3) and in-situ Chl-a. [B3-560 nm, B5-705 nm]

reflectance peak near 700 nm is important in measuring the chl-a concentration via satellite remote sensing. Gurlin [21] stated that a spectral range of 705 nm to 570 nm has potential in the estimation of chl-a concentration. The magnitude and reflectance peak shows the spectral signature (Fig. 2) that belongs to turbid water (Fig. 1). The weak absorption of chlorophyll-a concentration observes the reflectance peak near 580–590 nm. The reflectance peak near 705 nm shows strong absorption of chl-a concentration.

3.3. In-situ dataset

Field-based optical remote sensing data is collected by measuring the upwelling radiance (L_w) and downwelling irradiance (E_d). Water samples are collected at the depth of one meter below the

Table 1
Statistical of chl-a concentration for calibration and validation.

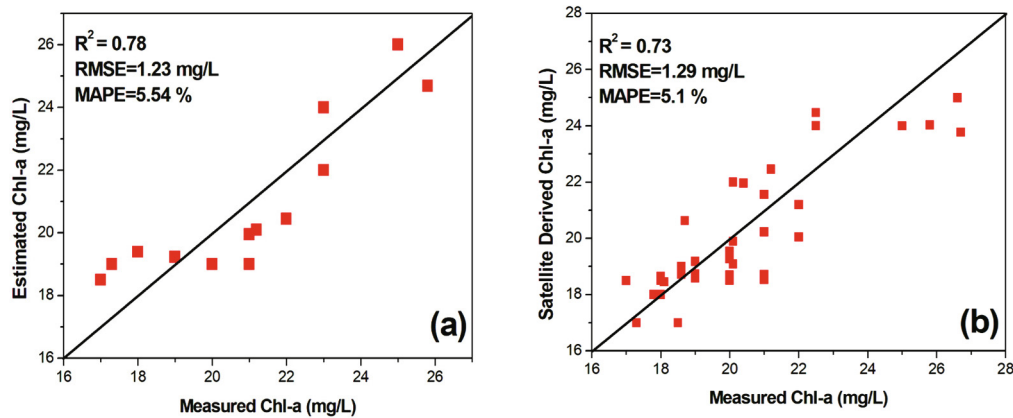
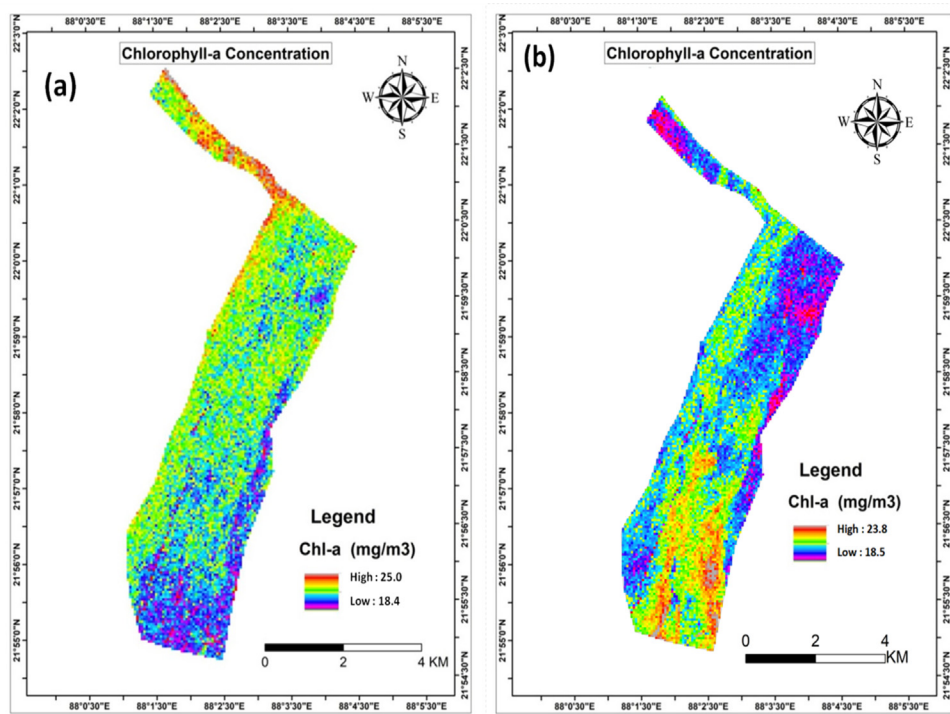
Datasets	No. of total data	Min. Chl-a (mg/m ³)	Max. Chl-a (mg/ m ³)	Avg. Chl-a (mg/ m ³)	S.D Chl-a (mg/ m ³)	C.V (%)
Calibration	26	17.8	26.7	20.5	2.57	0.125
In-situ Validation	14	17.0	25.8	20.9	2.70	0.130
Satellite validation	40	18.4	25.0	20.0	2.30	0.111

surface water at all sampling points. The in-situ data is collected on availability of satellite data with clear sky condition. To minimize in-situ data error and water turbulence during data collection, the boat is turned off. Sat-view is open software developed by a Sea-bird scientific company for the real-time acquisition of in-situ hyperspectral data. The in-situ remote sensing reflectance is calculated by equation (3).

Table 2

Calibration and validation results of chl-a concentration.

Calibration					Validation			
Model	R ²	a	b	c	Model	R ²	RMSE validation (mg/L)	MAPE validation (%)
Polynomial 2 band model	0.77	218.8	-415.7	215.9	In-situ validation	0.78	1.23	5.54
					Satellite validation	0.73	1.29	5.07

**Fig. 4.** In-situ validation (a) and satellite validation (b) between the measured and estimated chlorophyll-a concentration.**Fig. 5.** a & b Spatial distribution of chlorophyll-a concentration retrieved using Sentinel-2 MSI satellite data for 2018 and 2019.

work on developing the regional algorithm for retrieval of chl-a concentration in this study area. The developed algorithm is validated by statistical analysis. The statistical analysis shows that in-situ chl-a and in-situ remote sensing reflectance are well correlated with $R^2 = 0.77$. The in-situ validation shows good regression analysis with $R^2 = 0.78$, RMSE = 1.23 mg/m³, MAPE = 5.54% (Fig. 3).

For satellite validation, remote sensing reflectance is retrieved by Sentinel-2 satellite data [23]. The derived satellite R_{rs} are well correlated with in-situ chl-a is shown in Fig. 4(a and b). The correlation results show a good agreement between in-situ and satellite-derived chl-a with $R^2 = 0.73$, RMSE = 1.29 mg/m³, MAPE = 5.1%. The developed algorithm is used to create a spatial distribution map of chl-a concentration. In the study area, the observed chl-a concentration is maximum in the northern part and it gradually decreases when it reaches the downstream of the river. This is because the upstream of the study area is stressed due to the international ship route, human habitation and merging of two rivers [24]. The spatial distribution of chl-a concentration (Fig. 5(a and b) shows that it has the highest concentration in the estuarine region and the lowest in the coastal region [25]. Fig. 5

4. Conclusion

Satellite hyperspectral ocean colour radiometer was used to collect the spectral signature of Hooghly River. This is the first approach made in this study area, developing the regional algorithm for retrieving of chl-a concentration. The in-situ dataset is arranged in ascending order from low to high values. Among them, 65% of the data is used for calibration and 35% of the data used for in-situ validation by the statistical method called systematic sampling. The developed regional algorithm is applied to Sentinel-2 MSI data for satellite validation. The case-2 regional coastal colour (C2RCC) atmospheric correction tool is applied to Sentinel-2 data. SNAP is open-source software which is used to retrieve satellite-based remote sensing reflectance. The satellite-based R_{rs} is applied to the developed model to retrieve chl-a concentration. The satellite validation shows a strong correlation between in-situ chl-a and satellite-derived chl-a concentration ($R^2 = 0.73$). The band ratio of 705/560 is suitable for retrieval of chl-a concentration in the study area with Sentinel-2 satellite data. Finally, this research work will aid in monitoring the spatial and temporal variation of chl-a concentration in the Hooghly River.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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